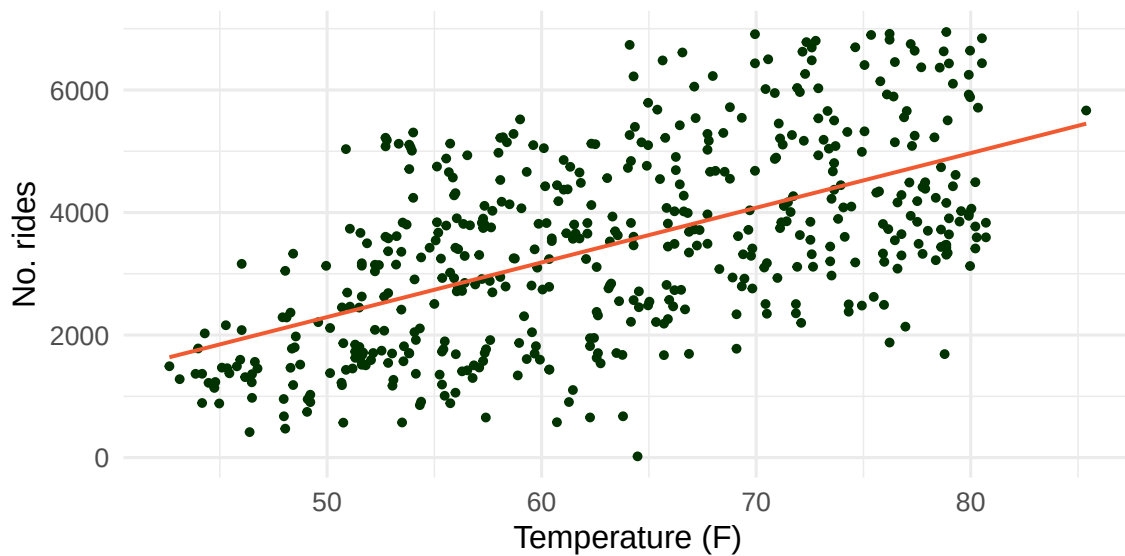


Day 09: Intro to Multiple Linear Regression

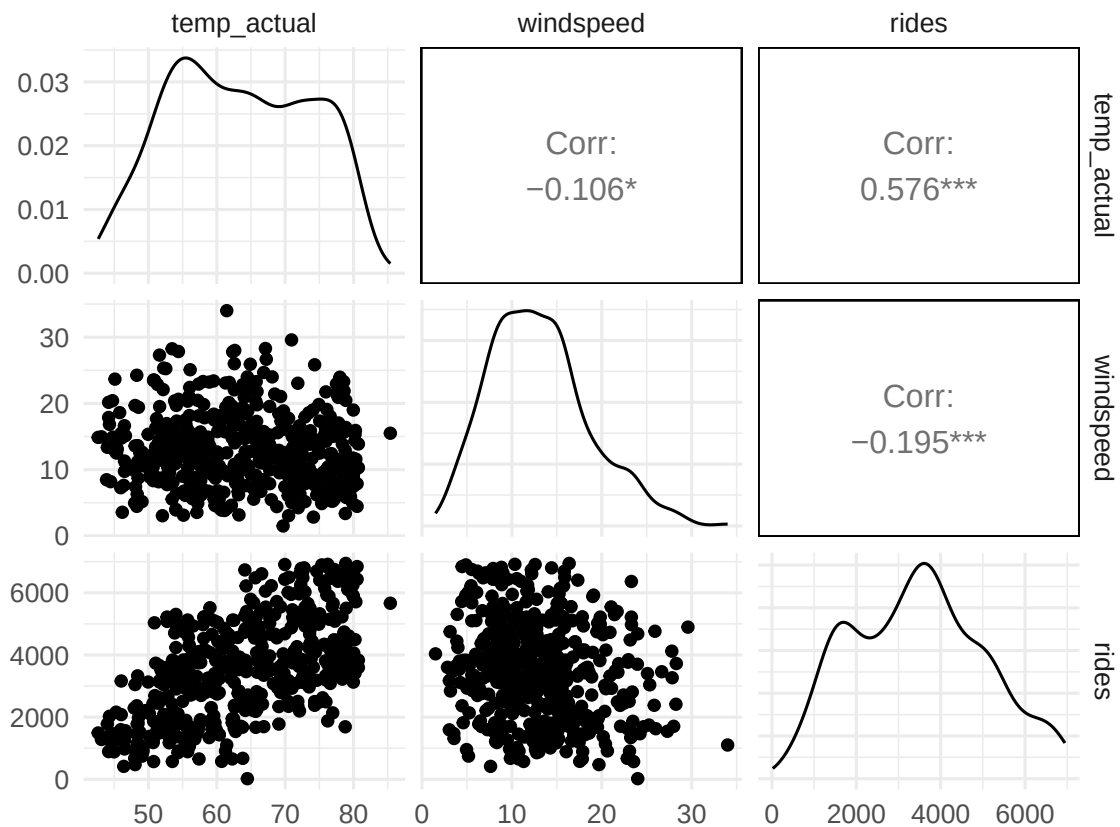
Back to bikes: one predictor

$$\widehat{\text{rides}} = -2168.77 + 89.23(\text{temp_actual})$$



$R^2 = 0.332$ – temperature alone leaves a lot of variation in ridership unexplained. What if we add a second predictor?

Pairs plot and correlation matrix



```
bikes |>
  select(temp_actual, windspeed, rides) |>
  cor() |> round(3)
```

	temp_actual	windspeed	rides
temp_actual	1.000	-0.106	0.576
windspeed	-0.106	1.000	-0.195
rides	0.576	-0.195	1.000

Multiple Linear Regression (MLR)

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

This model requires:

1. $\mu\{Y \mid x_1, x_2, \dots, x_p\}$ is a _____ function of the predictors
2. The error terms are _____
3. The error terms follow a Normal distribution with mean 0 and variance σ^2

Fitting the model

Procedure: least squares estimation chooses the coefficients to minimize

$$\text{SSE} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2, \quad \text{then } \hat{\sigma} = \sqrt{\frac{\text{SSE}}{n - (p + 1)}}$$

```
bikes_mlr <- lm(rides ~ temp_actual + windspeed, data = bikes)
coef(bikes_mlr)
```

```
(Intercept) temp_actual  windspeed
-1525.53822    87.01789   -38.62817
```

Interpreting $\hat{\beta}_0$

```
(Intercept) temp_actual  windspeed
-1525.538    87.018    -38.628
```

$\hat{\beta}_0$ is the expected value of the response when _____. Here: the expected number of rides is about -1525.5 when temperature = 0 and windspeed = 0 – not a meaningful day in this dataset!

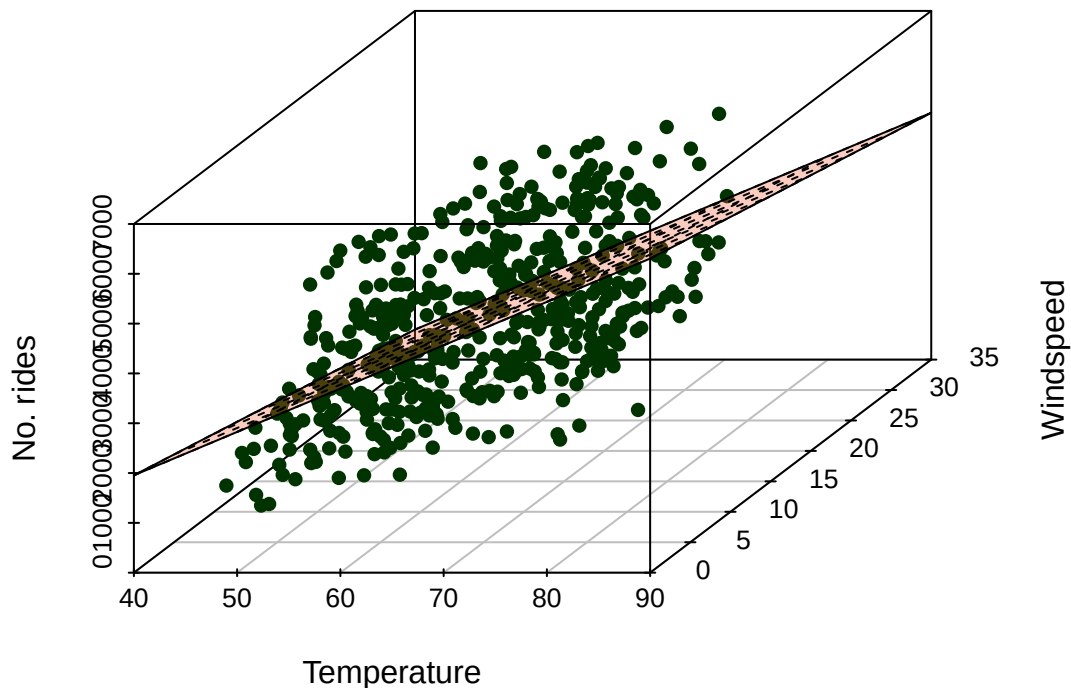
Interpreting $\hat{\beta}_j$

$\hat{\beta}_j$ is the expected change in the response for a 1-unit increase in x_j , when _____.

- We expect 87.02 *additional* rides for every 1 degree increase in actual temperature, holding wind-speed constant.
- We expect 38.63 *fewer* rides for every 1 MPH increase in windspeed, holding temperature constant.

Geometric interpretation: the MLR “line” is a plane

With two predictors, the fitted model is no longer a line but a **plane** through 3-dimensional space:



Centering predictor variables

Idea: recenter each predictor at its own mean, so an intercept of “all predictors = 0” becomes an intercept of “all predictors at their _____” – a value that actually occurs in the data.

```
bikes <- bikes |>
  mutate(
    temp_centered = temp_actual - mean(temp_actual, na.rm = TRUE),
    wind_centered = windspeed - mean(windspeed, na.rm = TRUE)
  )
```

Centering shifts the mean to 0 but leaves the spread alone:

mean_orig	mean_centered	sd_orig	sd_centered
63.33	0	10.17	10.17

Interpreting the centered model

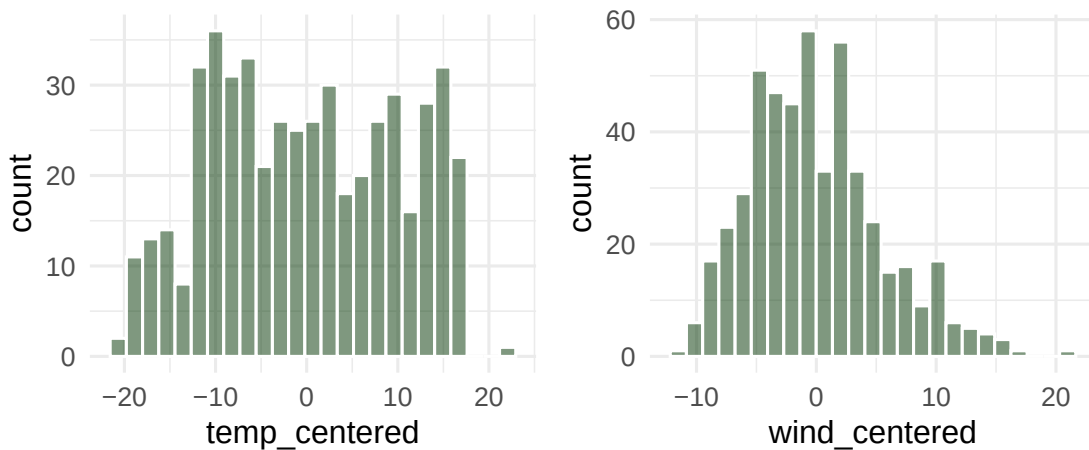
(Intercept) temp_centered wind_centered

3482.214 87.018 -38.628

- $\hat{\beta}_0$: the expected number of rides is about 3482 on an *average* temperature and windspeed day – much more interpretable than before.
- $\hat{\beta}_j$'s **don't change** – they still describe the expected change for a 1-unit increase in the original units of x_j , holding the other predictor constant.

Standardizing predictor variables

Idea: 1 MPH of windspeed isn't comparable to 1°F of temperature, so comparing raw slopes across predictors can be misleading. Standardize each predictor to have mean 0 and SD 1:



```
bikes <- bikes |>
mutate(
  temp_std = (temp_actual - mean(temp_actual, na.rm = TRUE)) / sd(temp_actual, na.rm = TRUE),
  wind_std = (windspeed - mean(windspeed, na.rm = TRUE)) / sd(windspeed, na.rm = TRUE)
)
```

mean_orig	mean_std	sd_orig	sd_std
63.33	0	10.17	1

Interpreting the standardized model

(Intercept) temp_std wind_std
3482.214 885.032 -213.560

- $\hat{\beta}_0$: expected rides on an average day is unchanged from the centered model (3482).
- $\hat{\beta}_j$: now the expected change in the response for a 1 _____ increase in x_j , holding the other predictor constant.

- We expect 885 *additional* rides for every 1 SD increase in temperature (10.2 degrees), holding windspeed constant.
- We expect 214 *fewer* rides for every 1 SD increase in windspeed (5.5 MPH), holding temperature constant.

 Note

Standardized coefficients put predictors measured in different units on a common scale, so their magnitudes become directly comparable.